

Gravity and the Armington Model

International Trade

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Gravity Equation

- Gravity equation of trade:

A strong empirical regularity stating that the value of trade between any pair of countries, i and j , is proportional to their economic size and inversely proportional to their geographic distance.

$$\text{Trade}_{ij} = \frac{(\text{GDP}_i)^\alpha (\text{GDP}_j)^\beta}{(\text{Distance}_{ij})^\gamma}$$

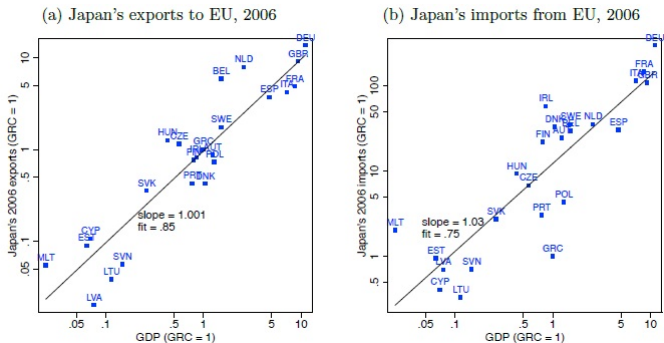
with $\alpha, \beta, \gamma \approx 1$.

Gravity Equation

- One of the simplest and most robust empirical regularities across economic sciences.
- Holds at different levels of aggregation, e.g., at the level of country, sectors, products, and firms.
- Holds for a wide range of flow variables where one can think of “size” and “distance”, e.g. migration flows, commuting flows, service offshoring, tourism, and knowledge transfers.

Size of origin and destination

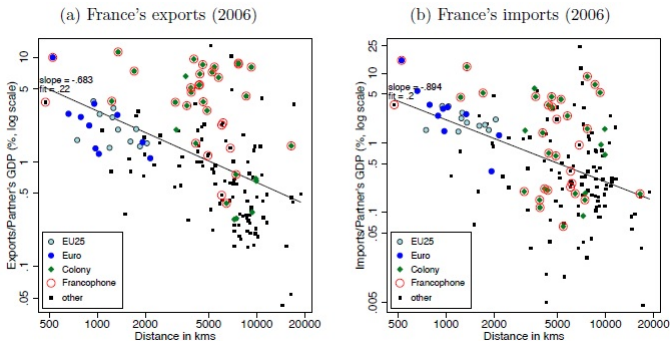
- Japan's exports and imports with the European Union.



Notes: Each EU member has about the same distance to Japan and applies the same trade policies on Japanese imports. Japan does not share a language, religion, currency or colonial history with any EU members either. Source: Head and Mayer, 2014.

Distance between origin and destination

- French exports and imports relative to partner's GDP against geographic and non-geographic distance.



Source: Head and Mayer, 2014

A specific example!

- Trade flows of crude oil

OLS estimation results for nonzero trade flows of crude oil, 2010 (Source: Farrokhi, 2020)

Dependent variable: log barrels of crude oil trade from i to n , $\log Q_{ni}$		
	(1)	(2)
log crude production of source, $\log Q_i$	1.041 (0.120)	
log total capacity of destination, $\log R_i$	1.074 (0.118)	
log distance, $\log dist_{ni}$	-1.059 (0.124)	-1.360 (0.159)
border	-1.102 (0.434)	-0.436 (0.475)
source FE	no	yes
destination FE	no	yes
# of observations	359	359
R^2	0.327	0.550

Notes: Standard errors are in parenthesis. "border" is a dummy that equals one if the source and the destination countries share a common border.

Gravity

- Gravity models yield a bilateral trade equation that can be expressed as:

$$X_{ij} = S_i M_j \phi_{ij}$$

- X_{ij} , trade value from exporter i to importer j
- S_i , exporter i 's characteristics that improve its exports to everywhere
- M_j , importer j 's characteristics that improve its imports from everywhere
- ϕ_{ij} , trade barriers from i to j

Naive Gravity

- Naive gravity representation

$$X_{ij} = Y_i^\alpha Y_j^\beta \phi_{ij}$$

- Leading to estimation regression of the form:

$$\ln X_{ij} = \alpha \ln Y_i + \beta \ln Y_j + \underbrace{\gamma \ln (\text{distance})_{ij}}_{\ln \phi_{ij}} + \varepsilon_{ij}$$

- Prior to Anderson and van Wincoop (2003), the above specification (potentially by including additional control variables) served as the basis of empirical analyses of gravity equations.

Structural Gravity

- Structural gravity:

$$X_{ij} = \frac{Y_i}{\Omega_i} \frac{X_j}{\Phi_j} \delta_{ij}$$

- $Y_i = \sum_j X_{ij}$, Country i 's total sales (to all destinations, incl. itself)
- $X_j = \sum_i X_{ij}$, Country j 's total expenditures (from all origins, incl. itself)
- Ω_i, Φ_j , Multilateral resistance terms:

$$\Omega_i = \sum_n \frac{\delta_{ni} X_n}{\Phi_n}, \quad \Phi_j = \sum_n \frac{\delta_{jn} Y_n}{\Omega_n}$$

- At this point, this specification may seem somewhat out of the blue. It is, however, called “structural” because it is grounded in a theoretical structure, as we show later in these slides.

Estimates of gravity equation of trade

Table 3.4 Estimates of Typical Gravity Variables

Estimates:	All Gravity				Structural Gravity			
	Median	Mean	s.d.	#	Median	Mean	s.d.	#
Origin GDP	.97	.98	.42	700	.86	.74	.45	31
Destination GDP	.85	.84	.28	671	.67	.58	.41	29
Distance	−.89	−.93	.4	1835	−1.14	−1.1	.41	328
Contiguity	.49	.53	.57	1066	.52	.66	.65	266
Common language	.49	.54	.44	680	.33	.39	.29	205
Colonial link	.91	.92	.61	147	.84	.75	.49	60
RTA/FTA	.47	.59	.5	257	.28	.36	.42	108
EU	.23	.14	.56	329	.19	.16	.5	26
NAFTA	.39	.43	.67	94	.53	.76	.64	17
Common currency	.87	.79	.48	104	.98	.86	.39	37
Home	1.93	1.96	1.28	279	1.55	1.9	1.68	71

Notes: The number of estimates is 2508, obtained from 159 papers. Structural gravity refers here to some use of country fixed effects or ratio-type method.

Source: Head and Mayer, 2014

Theory: Armington Model of International Trade

Road map

- We begin with the Armington model of trade to highlight the general equilibrium (GE) forces behind gravity.
 - The Armington (1969) assumption means that goods are differentiated by their origin of supply.
 - Like a “cheese economy” where you have preferences over French, Swiss and American cheese!
 - This is the simplest model to generate a gravity equation. (Anderson, 1979, Anderson and van Wincoop, 2003)
 - Useful to understand many other (more complicated) models of trade, particularly the ones that generate a structural gravity equation.

Setup

- Static
- N countries
- Each country i is populated by an exogenous measure L_i of homogeneous workers.
- Labor productivity in i is exogenously given by a_i .
- Each country produces a different variety of a good. (Armington assumption)
- “Iceberg trade costs”: in order for one unit of good that is produced in origin i to be arrived in destination n , origin i must ship d_{ij} units.
 - A fraction $d_{ij} - 1$ melts on its way from i to j .
 - $d_{ij} \geq 1$, $d_{ii} = 1$, and $d_{in}d_{nj} \geq d_{ij}$

Demand: CES

- Constant Elasticity of Substitution (CES) preferences:

$$U_j = \left[\sum_{i=1}^N b_{ij}^{\frac{1}{\sigma}} q_{ij}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

- b_{ij} : demand shifter
- q_{ij} : quantity of the good from origin i that arrives at destination j .
- $\sigma > 0$: elasticity of substitution between varieties

Demand

- Representative consumer in j maximizes her utility:

$$\max_{\{q_{ij} \geq 0\}_{j=1}^N} \left[\sum_{i=1}^N b_{ij}^{\frac{1}{\sigma}} q_{ij}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad \text{subject to} \quad \sum_{i=1}^N p_{ij} q_{ij} \leq X_j$$

— $p_{ij} = p_i d_{ij}$: price of good i in j , X_j : total expenditures of country j

- Lagrangian:

$$\mathcal{L} = \left[\sum_{i=1}^N b_{ij}^{\frac{1}{\sigma}} q_{ij}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} - \lambda \left(\sum_{i=1}^N [p_{ij} q_{ij}] - X_j \right)$$

- F.O.C.s (assuming an interior solution)

$$\frac{\partial \mathcal{L}}{\partial q_{ij}} = 0 : \quad b_{ij}^{\frac{1}{\sigma}} q_{ij}^{\frac{-1}{\sigma}} \left[\sum_{j=1}^N b_{ij}^{\frac{1}{\sigma}} q_{ij}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{1}{\sigma-1}} = \lambda p_{ij}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0 : \quad \sum_{i=1}^N p_{ij} q_{ij} = X_j$$

Demand

- Using the F.O.C.s,

$$q_{ij} = \lambda^{-\sigma} \left[\sum_{j=1}^N b_{ij}^{\frac{1}{\sigma}} q_{ij}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} b_{ij} p_{ij}^{-\sigma}$$

$$X_j = \sum_{i=1}^N p_{ij} q_{ij} = \lambda^{-\sigma} \left[\sum_{j=1}^N b_{ij}^{\frac{1}{\sigma}} q_{ij}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \sum_{i=1}^N b_{ij} p_{ij}^{1-\sigma}$$

- Replace $\lambda^{-\sigma} \left[\sum_{j=1}^N b_{ij}^{\frac{1}{\sigma}} q_{ij}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$ from the second eq. to the first to solve for q_{ij} and X_{ij} ,

$$q_{ij} = \frac{b_{ij} p_{ij}^{-\sigma}}{P_j^{1-\sigma}} X_j \quad (1)$$

$$X_{ij} \equiv p_{ij} q_{ij} = b_{ij} \left(\frac{p_{ij}}{P_j} \right)^{1-\sigma} X_j \quad (2)$$

where P_j is the CES price index,

$$P_j = \left[\sum_{i=1}^N b_{ij} p_{ij}^{1-\sigma} \right]^{\frac{1}{1-\sigma}} \quad (3)$$

Supply and market structure

- Production function,

$$Q_i = a_i L_i \quad (4)$$

- Markets are perfectly competitive.
- Price equals marginal cost:

$$p_{ij} = p_i d_{ij}, \quad \text{where} \quad p_i = \frac{w_i}{a_i}. \quad (5)$$

Market Clearing and General Equilibrium

- Market clearing,

$$\text{(Goods MCC)} \quad X_j = \sum_i p_{ij} q_{ij} \quad (6)$$

$$\text{(Labor MCC)} \quad Y_i \equiv w_i L_i = \sum_j p_{ij} q_{ij} \quad (7)$$

- **Definition.** Given endowments $\{L_i\}$, and parameters of demand, supply and trade costs $\{\sigma, b_{ij}, a_i, d_{ij}\}$, a *general equilibrium* is a vector of wage rates $\{w_i\}$ such that:
 - trade quantities and values $\{q_{ij}, X_{ij}\}$ are given by equations 1 and 2,
 - prices $\{P_j, p_i, p_{ij}\}$ by equations 3 and 5,
 - output quantities by equation 4,
 - and national expenditures and income levels $\{X_j, Y_i\}$ by equations 6 and 7.

Gravity

- Replace p_{ij} from eq. (5) into the trade flow eq. (2),

$$X_{ij} = b_{ij} p_i^{1-\sigma} \frac{d_{ij}^{1-\sigma}}{P_j^{1-\sigma}} X_j, \quad p_i = \left(\frac{w_i}{a_i} \right)$$

- Use Labor MCC (7),

$$Y_i = \sum_j X_{ij} = p_i^{1-\sigma} \times \sum_j b_{ij} \frac{d_{ij}^{1-\sigma}}{P_j^{1-\sigma}} X_j, \Rightarrow p_i^{1-\sigma} = \frac{Y_i}{\Pi_i^{1-\sigma}} \quad (8)$$

where:

$$\Pi_i = \left[\sum_j \frac{b_{ij} d_{ij}^{1-\sigma}}{P_j^{1-\sigma}} X_j \right]^{\frac{1}{1-\sigma}} \quad (9)$$

- Replace it in X_{ij} delivers a gravity equation,

$$X_{ij} = \frac{Y_i}{\Pi_i^{1-\sigma}} \times \frac{X_j}{P_j^{1-\sigma}} \times b_{ij} d_{ij}^{1-\sigma} \quad (10)$$

Gravity

- Using Eqs (3) and (8), the price index can be equivalently expressed as:

$$P_j^{1-\sigma} = \sum_{i=1}^N b_{ij} p_{ij}^{1-\sigma} = \sum_{i=1}^N b_{ij} p_i^{1-\sigma} d_{ij}^{1-\sigma} = \sum_{i=1}^N b_{ij} \frac{Y_i}{\Pi_i^{1-\sigma}} d_{ij}^{1-\sigma}$$

- We have now derived a structural gravity equation:

$$\text{(gravity equation of trade)} \quad X_{ij} = \frac{Y_i}{\Pi_i^{1-\sigma}} \times \frac{X_j}{P_j^{1-\sigma}} \times b_{ij} d_{ij}^{1-\sigma}$$

$$\text{(total income)} \quad Y_i = \sum_j X_{ij}$$

$$\text{(total expenditure)} \quad X_j = \sum_i X_{ij}$$

$$\text{(outward multilateral resistance)} \quad \Pi_i^{1-\sigma} = \sum_j \frac{X_j}{P_j^{1-\sigma}} b_{ij} d_{ij}^{1-\sigma}$$

$$\text{(inward multilateral resistance)} \quad P_j^{1-\sigma} = \sum_{i=1}^N \frac{Y_i}{\Pi_i^{1-\sigma}} b_{ij} d_{ij}^{1-\sigma}$$

General Equilibrium

- Having derived the gravity equation, we can now characterize the general equilibrium compactly.
- Use $p_{ij} = d_{ij}w_i/a_i$, and replace equation (3) into (2), $X_{ij} = \pi_{ij}X_j$, where π_{ij} is the share of j 's expenditure on i (trade share),

$$\pi_{ij} = \frac{b_{ij}d_{ij}^{1-\sigma} (w_i/a_i)^{1-\sigma}}{\sum_{\ell=1}^N b_{\ell j}d_{\ell j}^{1-\sigma} (w_{\ell}/a_{\ell})^{1-\sigma}}$$

- Using labor MCC, $w_iL_i = \sum_{j=1}^N X_{ij}$, and the balance of budget, $X_j = w_jL_j$,

$$w_iL_i = \sum_{j=1}^N \pi_{ij}(\mathbf{w}; \mathbf{a}, \mathbf{b}, \mathbf{d}, \sigma) w_jL_j, \quad \text{for all } i = 1, \dots, N.$$

General Equilibrium

- Reproduce the equilibrium equations:

$$\begin{cases} (a) & w_i L_i = \sum_{j=1}^N \pi_{ij}(\mathbf{w}; \mathbf{a}, \mathbf{b}, \mathbf{d}, \sigma) w_j L_j, & \text{for all } i = 1, \dots, N \\ (b) & \pi_{ij}(\mathbf{w}; \mathbf{a}, \mathbf{b}, \mathbf{d}, \sigma) = \frac{b_{ij} d_{ij}^{1-\sigma} (w_i/a_i)^{1-\sigma}}{\sum_{\ell=1}^N b_{\ell j} d_{\ell j}^{1-\sigma} (w_{\ell}/a_{\ell})^{1-\sigma}} & \text{for all } i, j = 1, \dots, N \end{cases}$$

- In eq (a), LHS is labor supply and RHS is labor demand.
- GE wage rates, $\mathbf{w} = [w_i]_{i=1}^N$, solve eq (a), where $\pi_{ij}(\cdot)$ is given by eq (b) and $\mathbf{L} = [L_i]$, $\mathbf{a} = [a_i]$, $\mathbf{b} = [b_{ij}]$, and $\mathbf{d} = [d_{ij}]$ and σ are exogenously given.
- Note, labor productivities, taste parameters, trade costs, and the substitution elasticity, $(\mathbf{a}, \mathbf{b}, \mathbf{d}, \sigma)$, affect GE only through their impact on trade shares.

Main References

- Anderson, James E. and Eric van Wincoop, (2003) “Gravity with Gravitas: A Solution to the Border Puzzle”. American Economic Review
- Keith Head and Thierry Mayer, (2014). “Gravity Equations: Workhorse, Toolkit, and Cookbook”, Chapter 3 in Vol. 4 of the Handbook of International Economics.